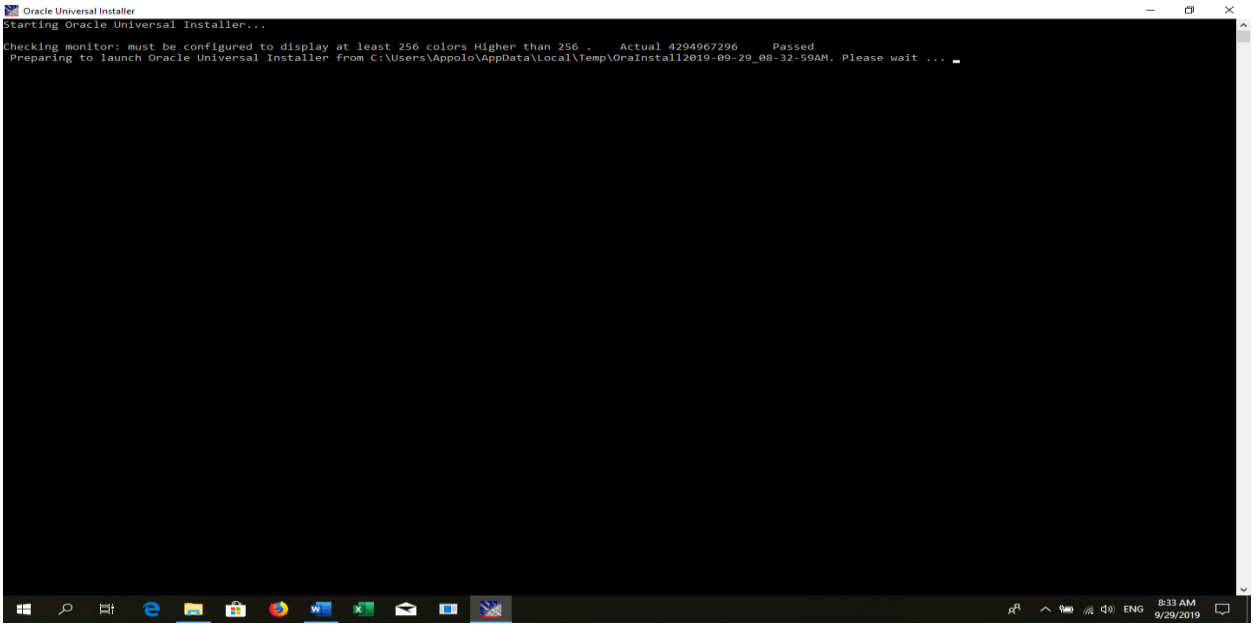
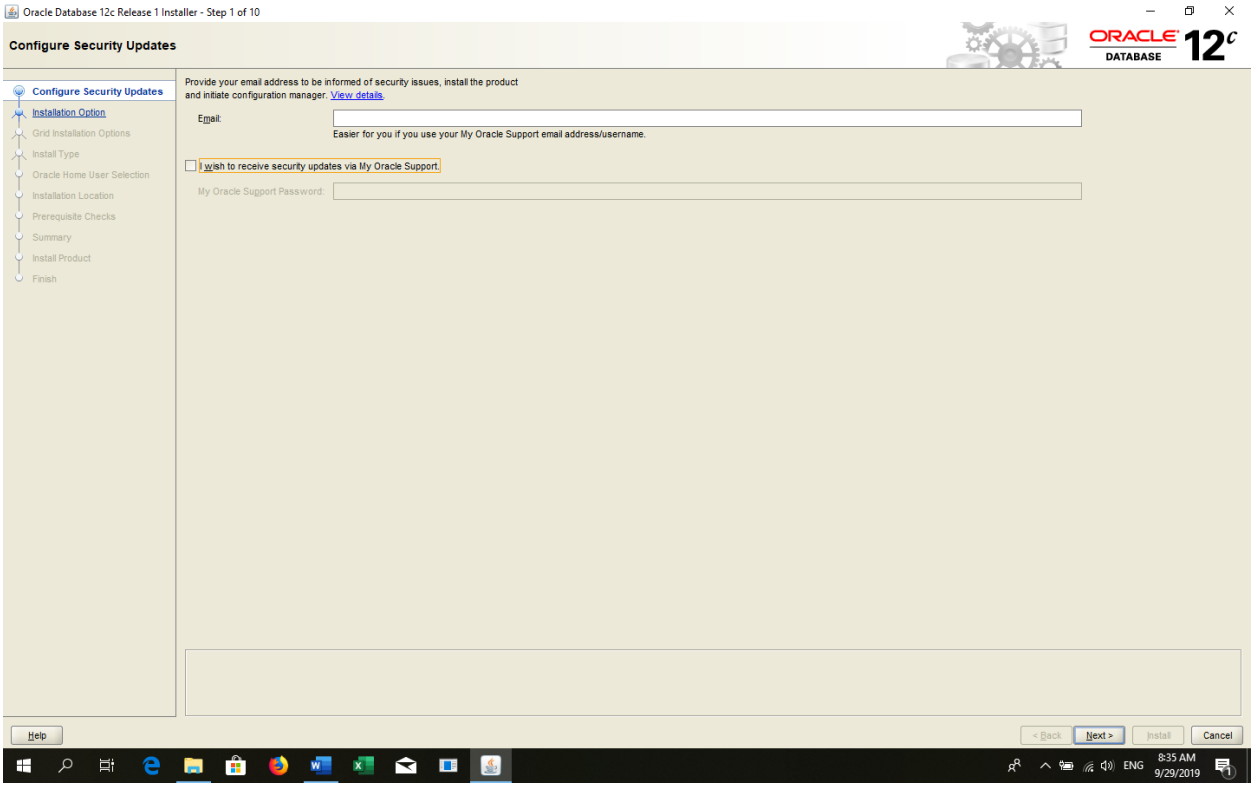
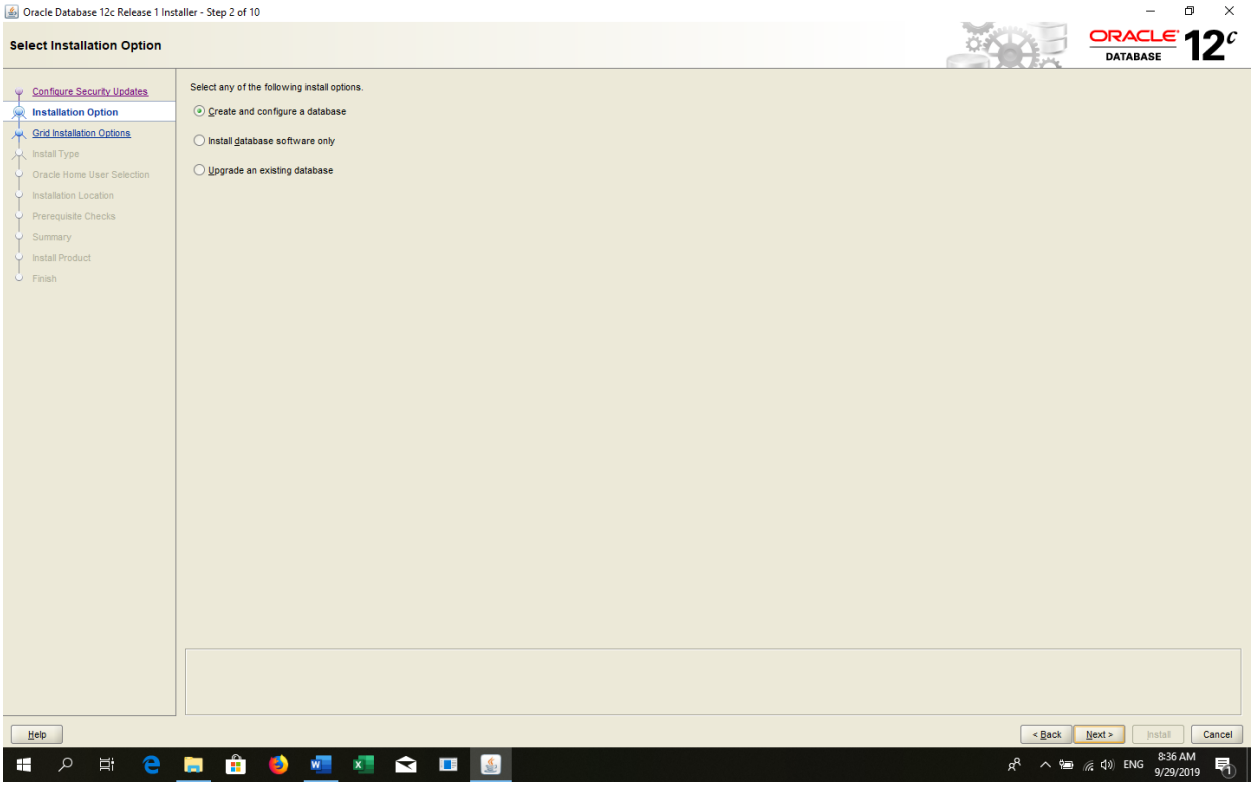
Crearea si administrarea unei baze de date Oracle

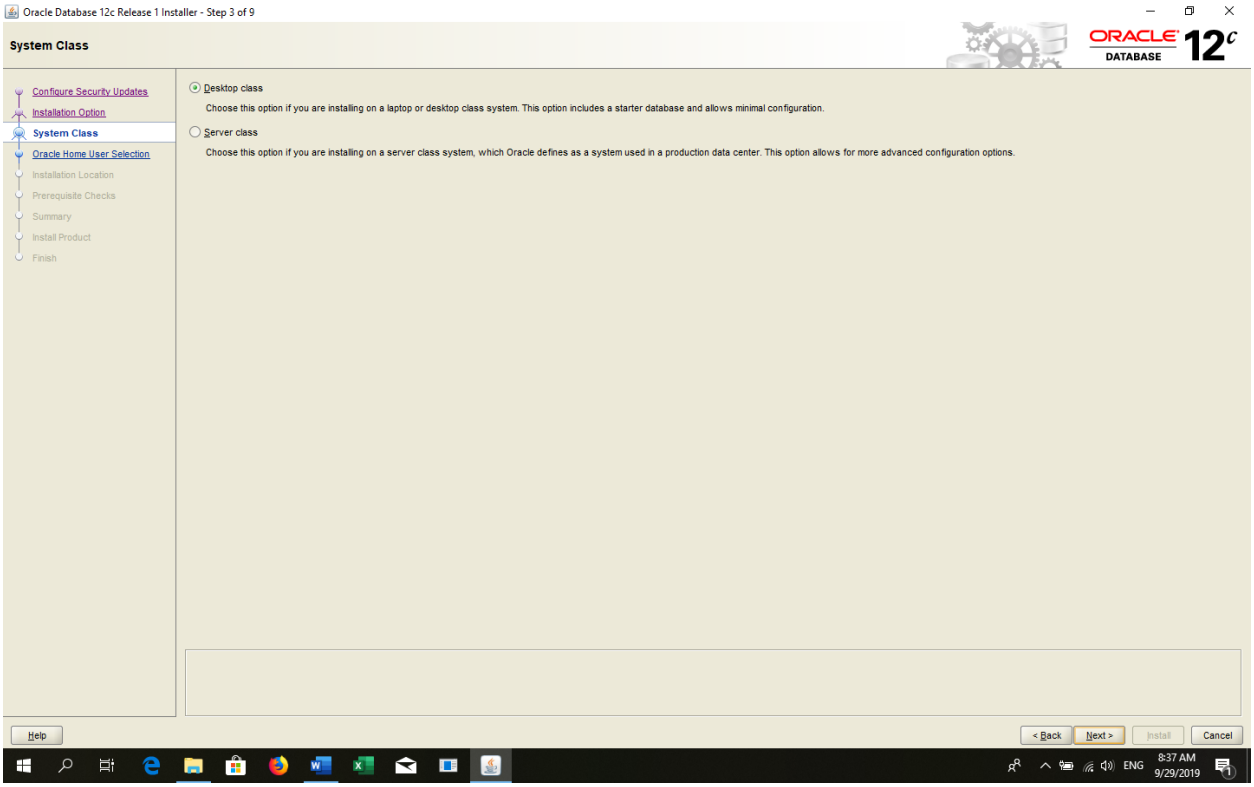
A. Instalarea bazei de date Oracle Database 12c

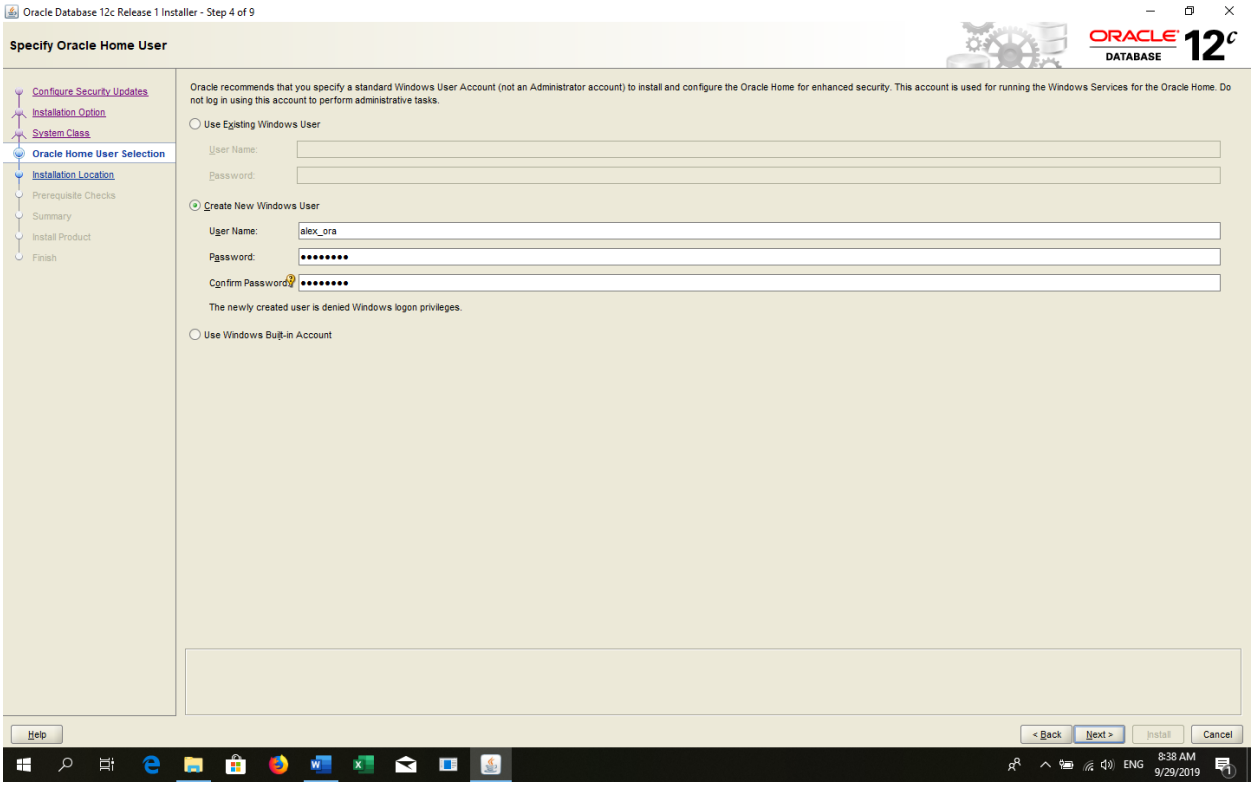
Run Setup as Administrator

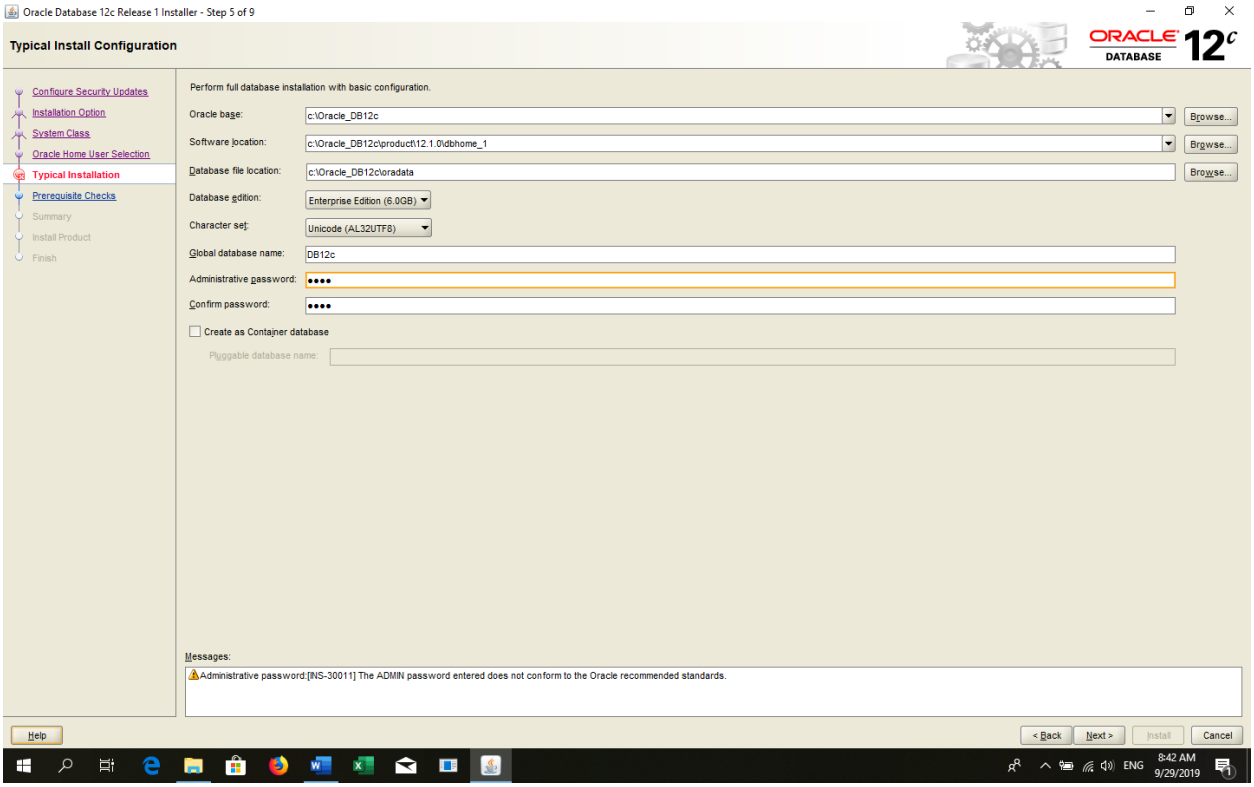


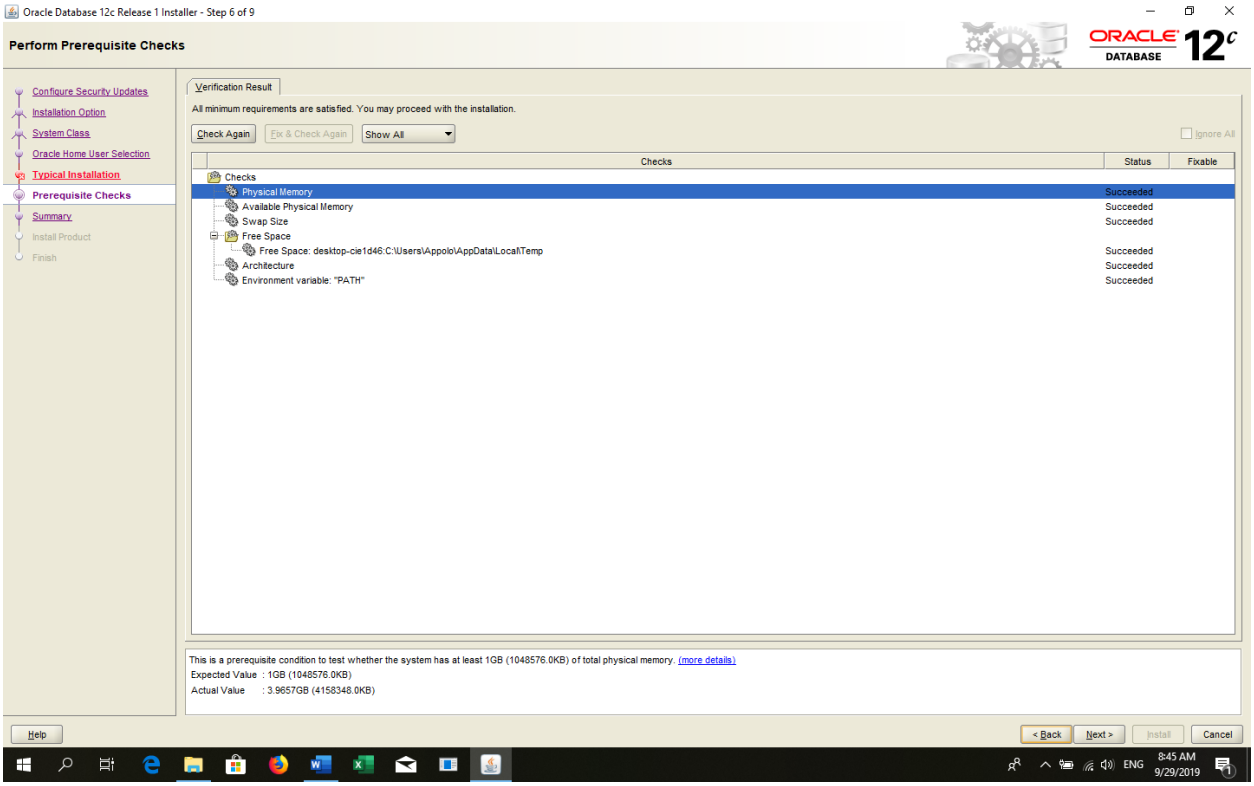


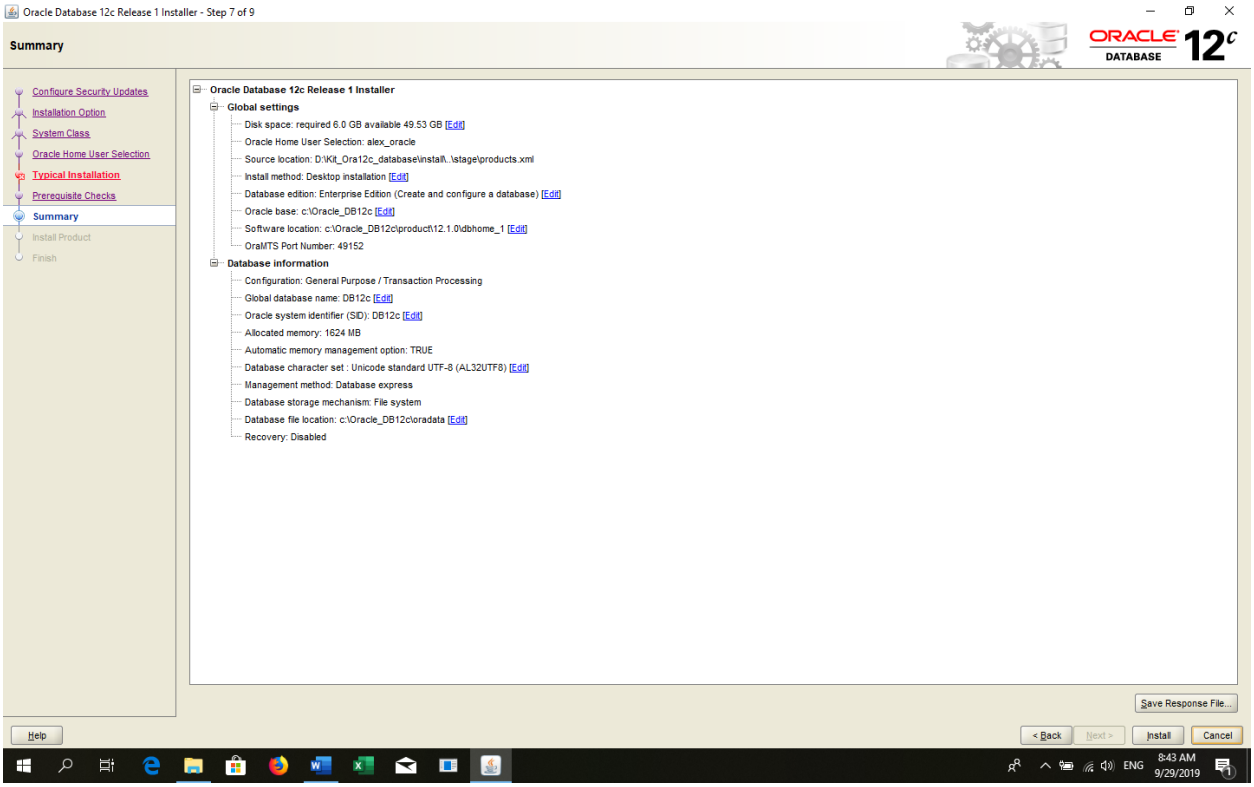

Oracle Database 12c Release 1 Installer - Step 2 of 10
Select Installation Option
ORACLE DATABASE 12c
Configure Security Updates
Installation Option
Grid Installation Options
Install Type
Oracle Home User Selection
Installation Location
Prerequisite Checks
Summary
Install Product
Finish
Select any of the following install options.
Create and configure a database
Install database software only
Upgrade an existing database
Help
< Back
Next >
Install
Cancel
ENG
8:36 AM
9/29/2019


Oracle Database 12c Release 1 Installer - Step 3 of 9
System Class
ORACLE DATABASE 12c
Configure Security Updates
Installation Option
System Class
Oracle Home User Selection
Installation Location
Prerequisite Checks
Summary
Install Product
Finish
Desktop class
Choose this option if you are installing on a laptop or desktop class system. This option includes a starter database and allows minimal configuration.
Server class
Choose this option if you are installing on a server class system, which Oracle defines as a system used in a production data center. This option allows for more advanced configuration options.
Help
< Back
Next >
Install
Cancel
ENG
8:37 AM
9/29/2019


Oracle Database 12c Release 1 Installer - Step 4 of 9
Specify Oracle Home User
ORACLE 12c DATABASE
Configure Security Updates
Installation Option
System Class
Oracle Home User Selection
Installation Location
Prerequisite Checks
Summary
Install Product
Finish
Oracle recommends that you specify a standard Windows User Account (not an Administrator account) to install and configure the Oracle Home for enhanced security. This account is used for running the Windows Services for the Oracle Home. Do not log in using this account to perform administrative tasks.
Use Existing Windows User
User Name:
Password:
Create New Windows User
User Name: alex_ora
Password: ••••••••
Confirm Password: ••••••••
The newly created user is denied Windows logon privileges.
Use Windows Built-in Account
Help
< Back
Next >
Install
Cancel
ENG
8:38 AM
9/29/2019


Oracle Database 12c Release 1 Installer - Step 5 of 9
Typical Install Configuration
ORACLE 12c DATABASE
Configure Security Updates
Installation Option
System Class
Oracle Home User Selection
Typical Installation
Prerequisite Checks
Summary
Install Product
Finish
Perform full database installation with basic configuration.
Oracle base: c:\Oracle_DB12c
Software location: c:\Oracle_DB12c\product\12.1.0\dbhome_1
Database file location: c:\Oracle_DB12c\oradata
Browse...
Database edition: Enterprise Edition (6.0GB)
Character set: Unicode (AL32UTF8)
Global database name: DB12c
Administrative password: ••••
Confirm password: ••••
Create as Container database
Pluggable database name:
Messages:
Administrative password:[INS-30011] The ADMIN password entered does not conform to the Oracle recommended standards.
Help
< Back
Next >
Install
Cancel
ENG
8:42 AM
9/29/2019


Oracle Database 12c Release 1 Installer - Step 6 of 9
Perform Prerequisite Checks
ORACLE DATABASE 12c
Configure Security Updates
Installation Option
System Class
Oracle Home User Selection
Typical Installation
Prerequisite Checks
Summary
Install Product
Finish
Verification Result
All minimum requirements are satisfied. You may proceed with the installation.
Check Again
Fix & Check Again
Show All
Ignore All
Checks
Status
Fixable
Checks
Physical Memory
Succeeded
Available Physical Memory
Succeeded
Swap Size
Succeeded
Free Space
Free Space: desktop-cie1d46:C:\Users\Appolo\AppData\Local\Temp
Succeeded
Architecture
Succeeded
Environment variable: "PATH"
Succeeded
This is a prerequisite condition to test whether the system has at least 1GB (1048576.0KB) of total physical memory. (more details)
Expected Value : 1GB (1048576.0KB)
Actual Value : 3.9657GB (4158348.0KB)
Help
< Back
Next >
Install
Cancel
ENG
8:45 AM
9/29/2019


Oracle Database 12c Release 1 Installer - Step 7 of 9
Summary
ORACLE DATABASE 12c
Configure Security Updates
Installation Option
System Class
Oracle Home User Selection
Typical Installation
Prerequisite Checks
Summary
Install Product
Finish
Oracle Database 12c Release 1 Installer
Global settings
Disk space: required 6.0 GB available 49.53 GB [Edit]
Oracle Home User Selection: alex_oracle
Source location: D:\Kit_Ora12c_database\install\..\stage\products.xml
Install method: Desktop installation [Edit]
Database edition: Enterprise Edition (Create and configure a database) [Edit]
Oracle base: c:\Oracle_DB12c [Edit]
Software location: c:\Oracle_DB12c\product\12.1.0\dbhome_1 [Edit]
OraMTS Port Number: 49152
Database information
Configuration: General Purpose / Transaction Processing
Global database name: DB12c [Edit]
Oracle system identifier (SID): DB12c [Edit]
Allocated memory: 1624 MB
Automatic memory management option: TRUE
Database character set : Unicode standard UTF-8 (AL32UTF8) [Edit]
Management method: Database express
Database storage mechanism: File system
Database file location: c:\Oracle_DB12c\oradata [Edit]
Recovery: Disabled
Save Response File...
Help
< Back
Next >
Install
Cancel
ENG
8:43 AM
9/29/2019

B. Crearea unei baze de date prin comenzi manuale

Pas 1 : Conectarea la baza de date

Se conecteaza cu userul SYS si privilegiul SYSDBA.

Enter user-name: **sys@BD as sysdba**
Enter password: ****

Pas 2 : Se porneste baza de date in starea NOMOUNT

SQL> startup nomount;
ORACLE instance started.
Total System Global Area 21790532 bytes
Fixed Size 278340 bytes
Variable Size 16777216 bytes
Database Buffers 4194304 bytes
Redo Buffers 540672 bytes

Pas 3 : Se creeaza si se executa comanda CREATE DATABASE

Se editeaza un script care continue comanda CREATE DATABASE si se executa scriptul:
SQL> @e:\student\script1.sql

Exemplu:

Obs. Parametrii specificati in comanda CREATE DATABASE trebuie sa se mapeze pe parametrii bazei de date care a fost deja creata la instalarea sistemului de gestiune.

```
CREATE DATABASE UBD
USER SYS IDENTIFIED BY sys_password
USER SYSTEM IDENTIFIED BY system_password
LOGFILE
GROUP 1 ('C:\Oracle_12c\oradata\BD\REDO01.rdo') SIZE 100M,
GROUP 2 ('C:\Oracle_12c\oradata\BD\REDO02.rdo') SIZE 100M,
GROUP 3 ('C:\Oracle_12c\oradata\BD\REDO03.rdo') SIZE 100M
DATAFILE 'C:\Oracle_12c\oradata\BD\SYSTEM01.dbf' SIZE 325M REUSE
SYSAUX DATAFILE 'C:\Oracle_12c\oradata\BD\SYSAUX01.dbf' SIZE 325M REUSE
MAXLOGFILES 5
MAXLOGMEMBERS 5
MAXLOGHISTORY 1
MAXDATAFILES 100
```

```
CHARACTER SET US7ASCII
NATIONAL CHARACTER SET AL16UTF16
EXTENT MANAGEMENT LOCAL
DEFAULT TABLESPACE users
DATAFILE 'C:\Oracle_12c\oradata\BD\USERS01.dbf' SIZE 500M REUSE AUTOEXTEND
ON MAXSIZE UNLIMITED
DEFAULT TEMPORARY TABLESPACE temp
TEMPFILE 'C:\Oracle_12c\oradata\BD\TEMP01.dbf' SIZE 20M REUSE
UNDO TABLESPACE undotbs1
DATAFILE 'C:\Oracle_12c\oradata\BD\UNDOTBS01.dbf' SIZE 200M REUSE
AUTOEXTEND ON MAXSIZE UNLIMITED
```

Pas 4 : Se trece baza de date in starea OPEN

SQL> alter database open;
Database altered.

Pas 5 : Se executa scripturile catalog.sql si catproc.sql

• Aceste scripturi se executa dupa crearea bazei de date:
- **catalog.sql** - continue comenzi de creare a tabelelor, sinonimelor si view-urilor dinamice in dictionarul bazei de date;
- **catproc.sql** - contine comenzi de creare a package-urilor si procedurilor necesare limbajului PL/SQL in dictionarul bazei de date. Acest script mai contine si alte package-uri pentru alerte, replicari, diferite tipuri de obiecte, etc.

• Scripturile se executa in userul SYS cu privilegiul SYSDBA.
• Baza de date trebuie sa fie in starea OPEN.

SQL> @e:\student\catalog.sql
SQL> @e:\student\catproc.sql

C. Stergerea din dictionar a unei baze de date prin comenzi manuale

Pas 1 : Conectarea la baza de date

Se conecteaza cu userul SYS si privilegiul SYSDBA.

Pas 2 : Localizarea fisierelor de date, control si log

Aceste fisiere trebuie sterse manual dupa stergerea bazei de date.

SQL> select name from v$datafile;

NAME

C:\ORACLE_12C\ORADATA\BD\SYSTEM01.DBF
C:\ORACLE_12C\ORADATA\BD\SYSAUX01.DBF
C:\ORACLE_12C\ORADATA\BD\UNDOTBS01.DBF
C:\ORACLE_12C\ORADATA\BD\USERS01.DBF
C:\ORACLE_12C\ORADATA\BD\EXAMPLE01.DBF
C:\ORACLE_12C\ORADATA\BD\BI_IAS_OPSS.DBF
C:\ORACLE_12C\ORADATA\BD\BI_UMS.DBF
C:\ORACLE_12C\ORADATA\BD\BI_MDS.DBF
C:\ORACLE_12C\ORADATA\BD\BI_IAU.DBF
C:\ORACLE_12C\ORADATA\BD\BI_BIPLATFORM.DBF
C:\ORACLE_12C\ORADATA\BD\BI_WLSSERVICES.DBF
C:\ORACLE_12C\ORADATA\BD\BI_SVCTBL.DBF

12 rows selected.

SQL> select name from v$controlfile;
NAME

C:\ORACLE_12C\ORADATA\BD\CONTROL01.CTL
C:\ORACLE_12C\ORADATA\BD\CONTROL02.CTL

SQL> select member from v$logfile;
MEMBER

C:\ORACLE_12C\ORADATA\BD\REDO03.LOG

C:\ORACLE_12C\ORADATA\BD\REDO02.LOG

C:\ORACLE_12C\ORADATA\BD\REDO01.LOG

Pas 3 : Oprirea bazei de date

SQL> shutdown immediate;
Database closed.
Database dismounted.
ORACLE instance shut down.

Pas 4: Pornirea bazei de date in modul Exclusive

SQL> startup mount exclusive restrict;

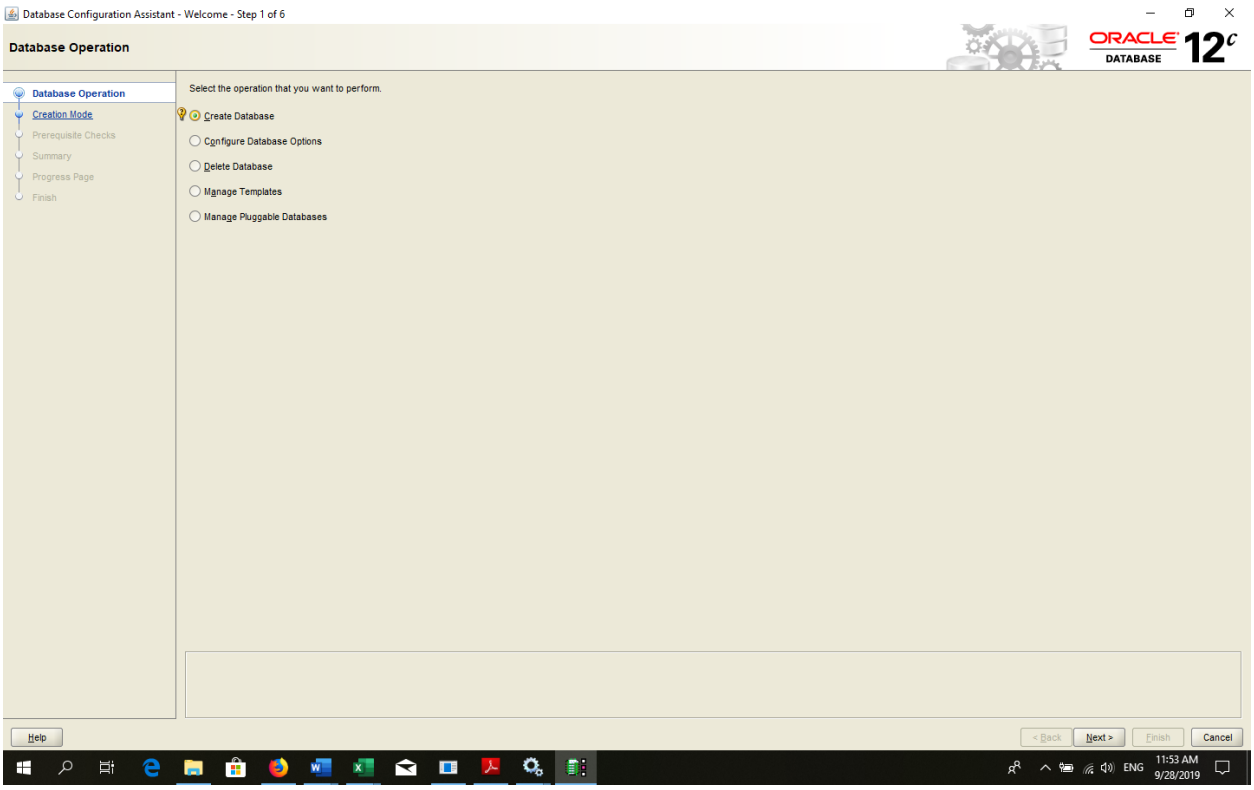
Pas 5: Stergerea bazei de date

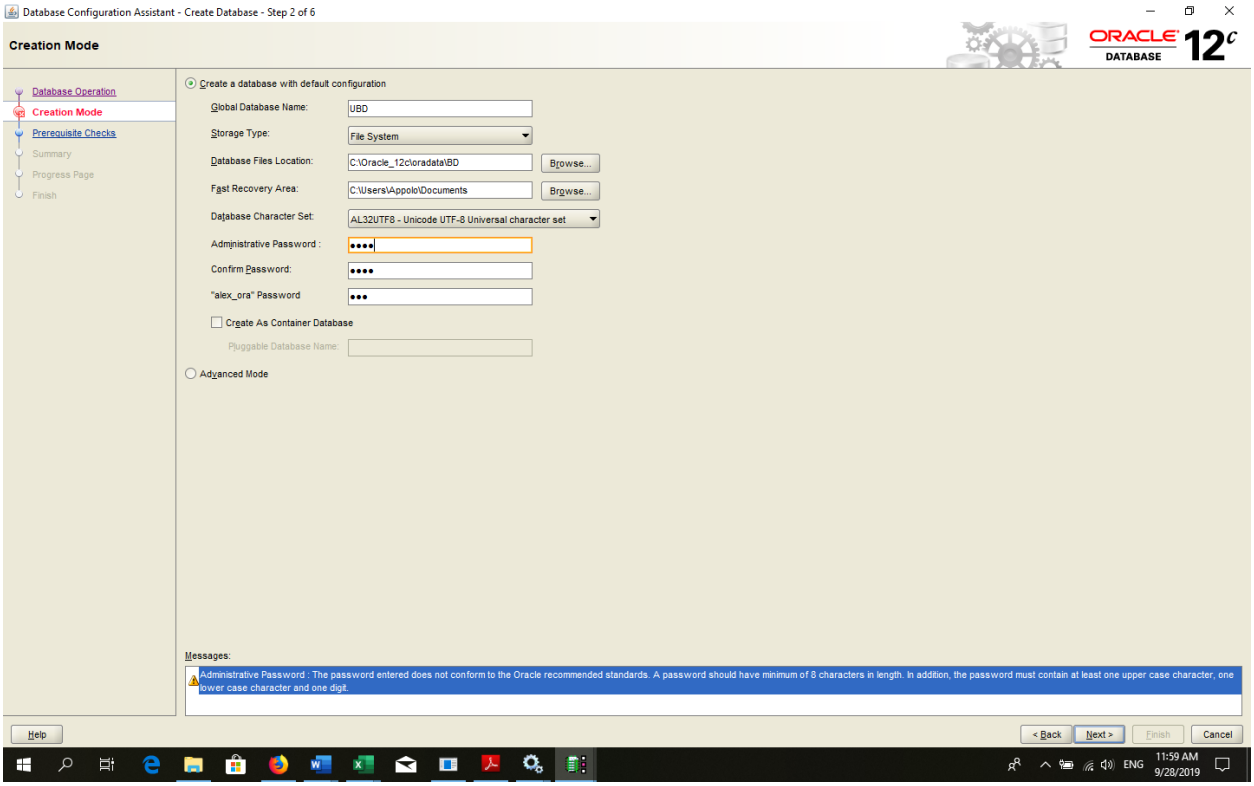
SQL> drop database;

Pas 6 : Verificari si stergeri de fisiere dupa stergerea bazei de date

1) Stergerea fisierelor de date, control si log cu comenzi din sistemul de operare;
2) Stergerea directoarelor devenite inutile;
3) Stergerea shortcut-urilor;
4) Modificarea scripturilor care fac referire la baza de date care a fost stearsa.

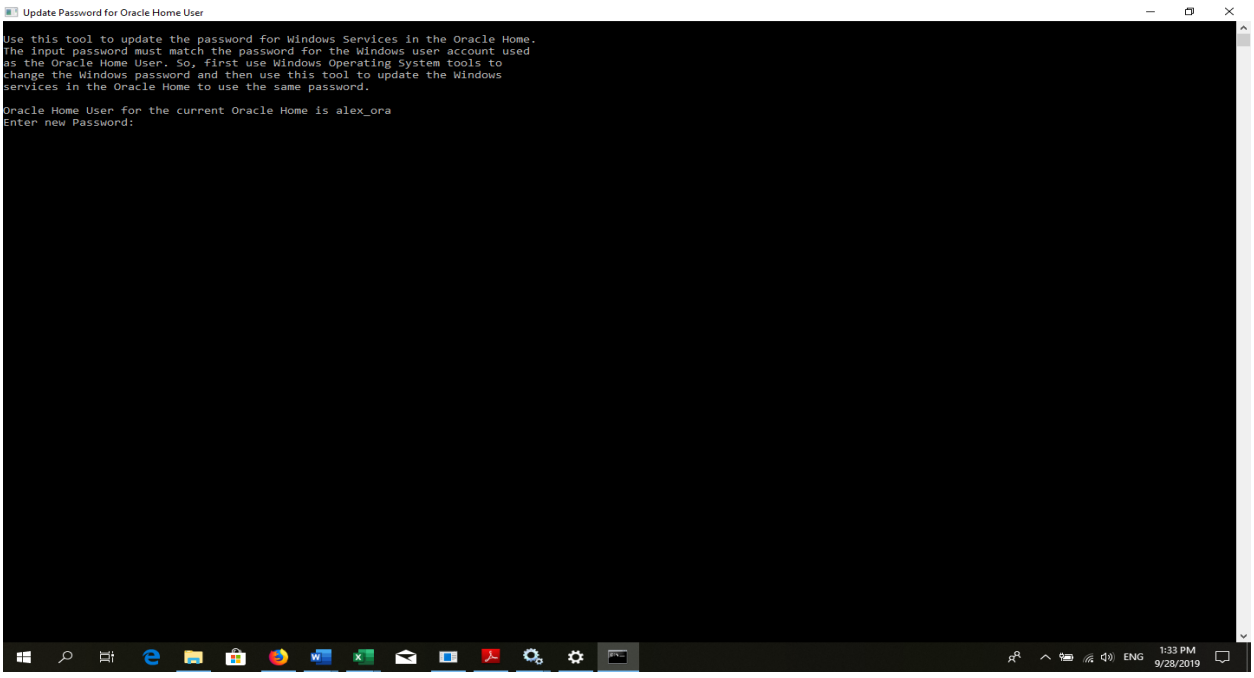
D. Crearea bazei de date cu utilitarul Database Configuration Assistant

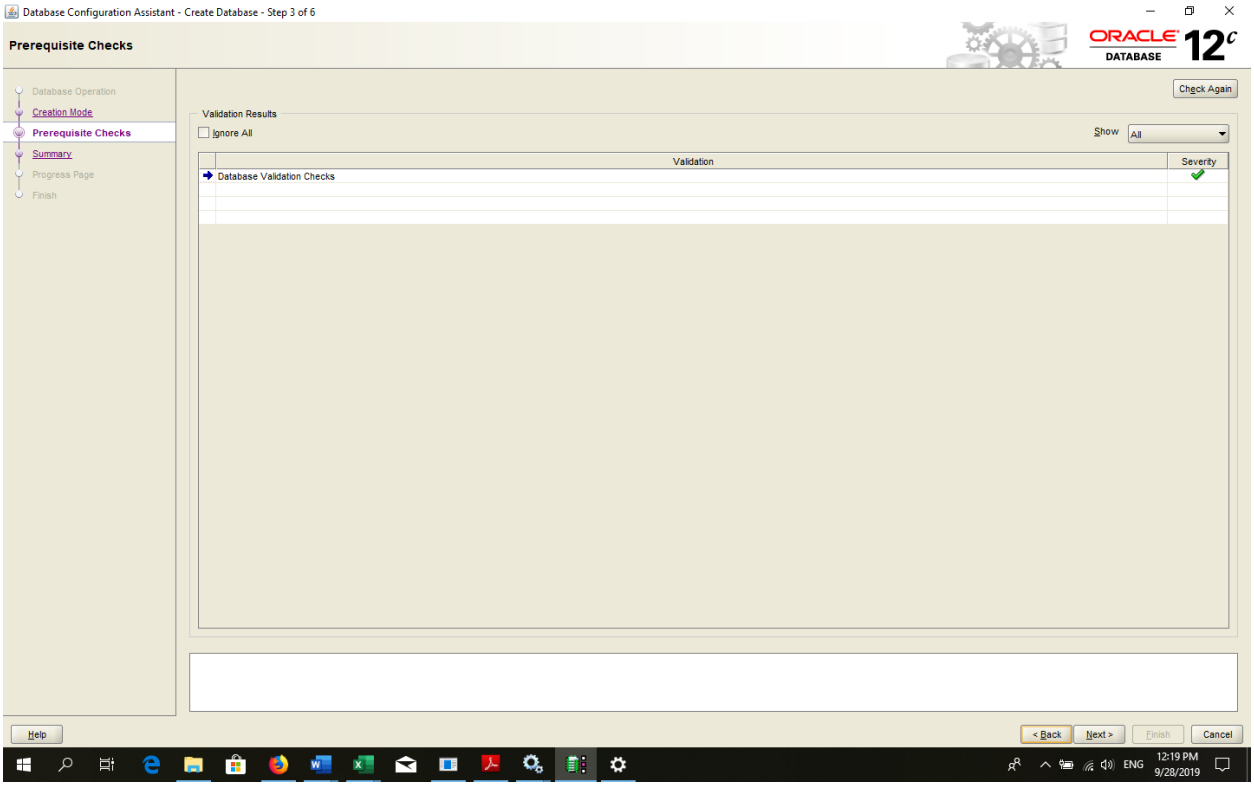


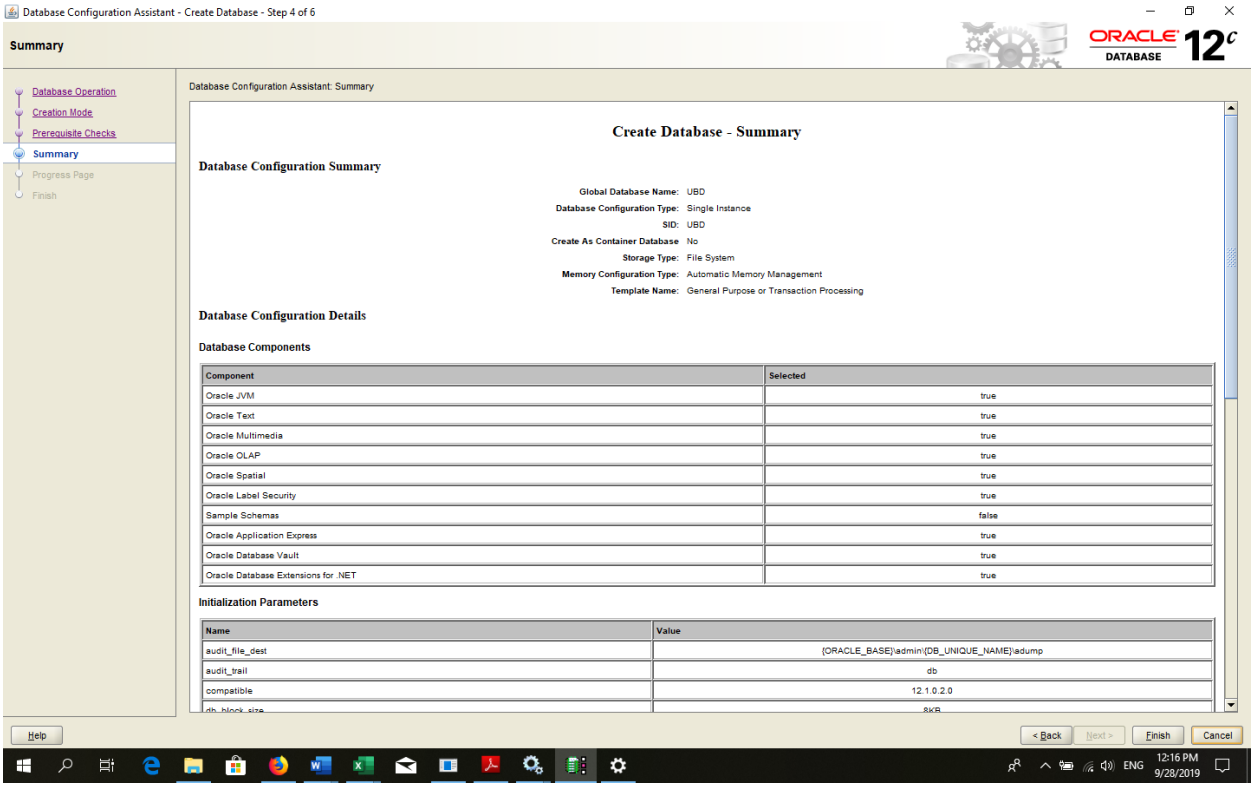


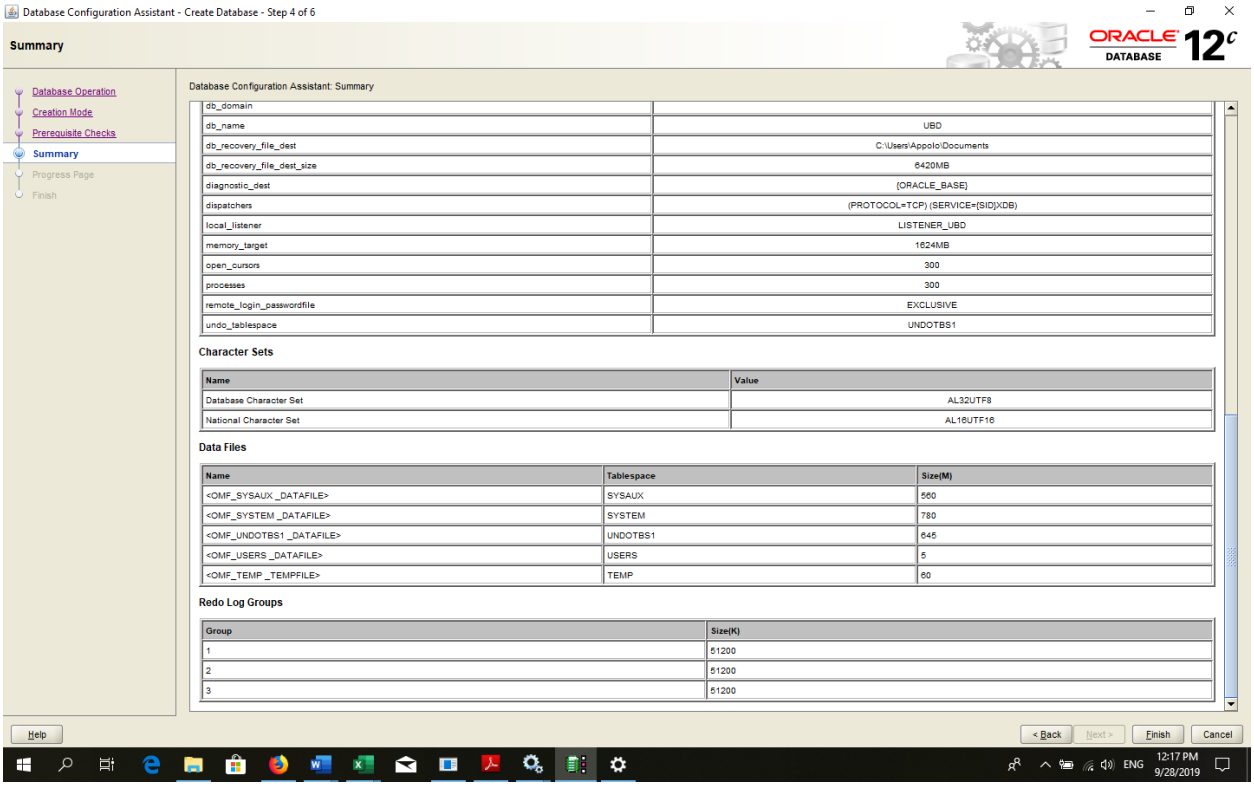
Obs:

- Se poate opta pentru crearea unui nou user extern pentru administrarea bazei de date.
- Parola pentru Oracle User se poate schimba din START => Oracle_Home1 => Update Password for Oracle Home User :

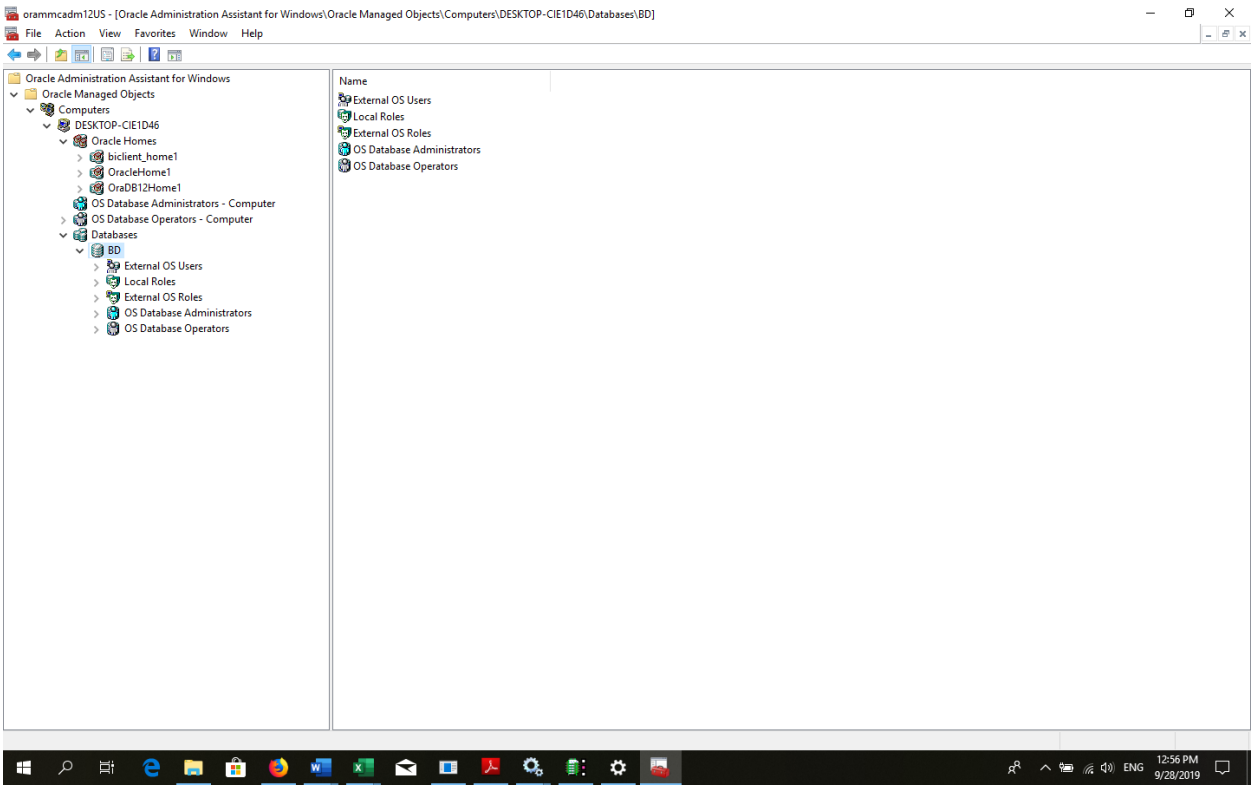



Database Configuration Assistant - Create Database - Step 3 of 6
Prerequisite Checks
ORACLE DATABASE 12c
Database Operation
Creation Mode
Prerequisite Checks
Summary
Progress Page
Finish
Check Again
Validation Results
Ignore All
Show
All
Validation
Severity
Database Validation Checks
Help
< Back
Next >
Finish
Cancel
ENG
12:19 PM
9/28/2019

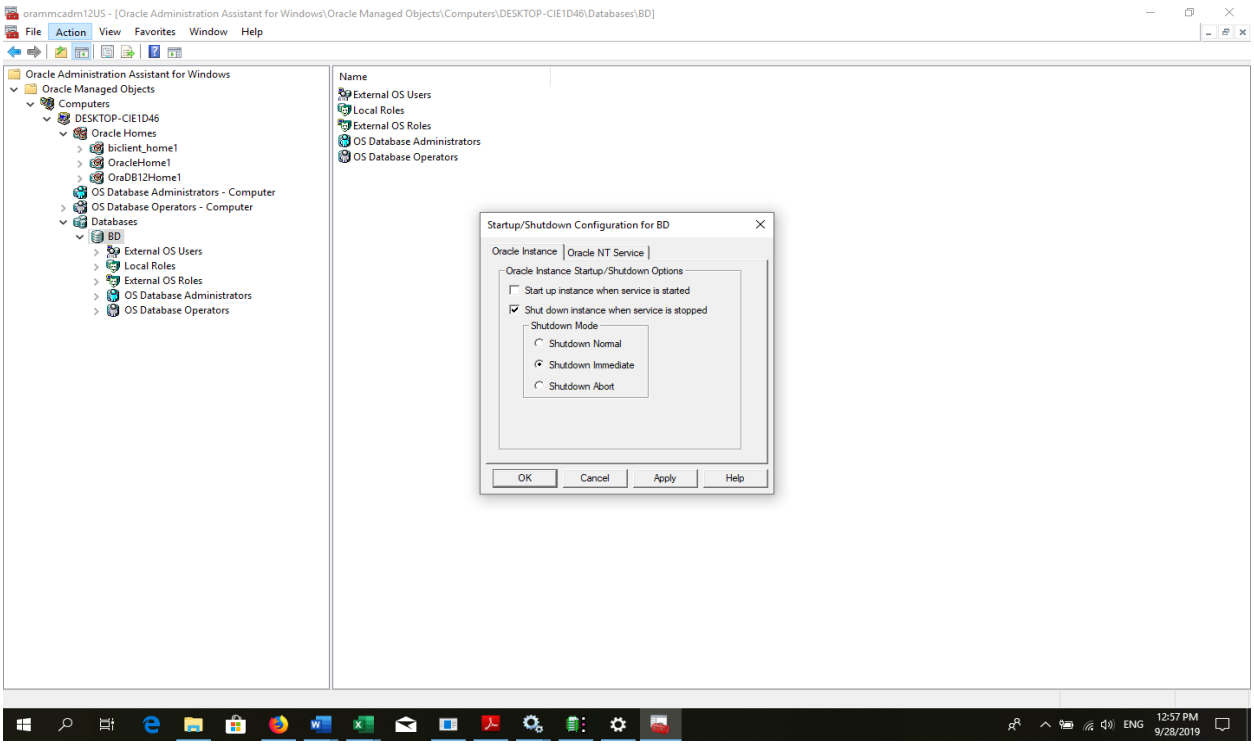

Database Configuration Assistant - Create Database - Step 4 of 6
Summary
ORACLE DATABASE 12c
Database Operation
Creation Mode
Prerequisite Checks
Summary
Progress Page
Finish
Database Configuration Assistant: Summary
Create Database - Summary
Database Configuration Summary
Global Database Name: UBD
Database Configuration Type: Single Instance
SID: UBD
Create As Container Database No
Storage Type: File System
Memory Configuration Type: Automatic Memory Management
Template Name: General Purpose or Transaction Processing
Database Configuration Details
Database Components
Component
Selected
Oracle JVM
true
Oracle Text
true
Oracle Multimedia
true
Oracle OLAP
true
Oracle Spatial
true
Oracle Label Security
true
Sample Schemas
false
Oracle Application Express
true
Oracle Database Vault
true
Oracle Database Extensions for .NET
true
Initialization Parameters
Name
Value
audit_file_dest
{ORACLE_BASE}\admin\{DB_UNIQUE_NAME}\adump
audit_trail
db
compatible
12.1.0.2.0
Help
< Back
Next >
Finish
Cancel
ENG
12:16 PM
9/28/2019



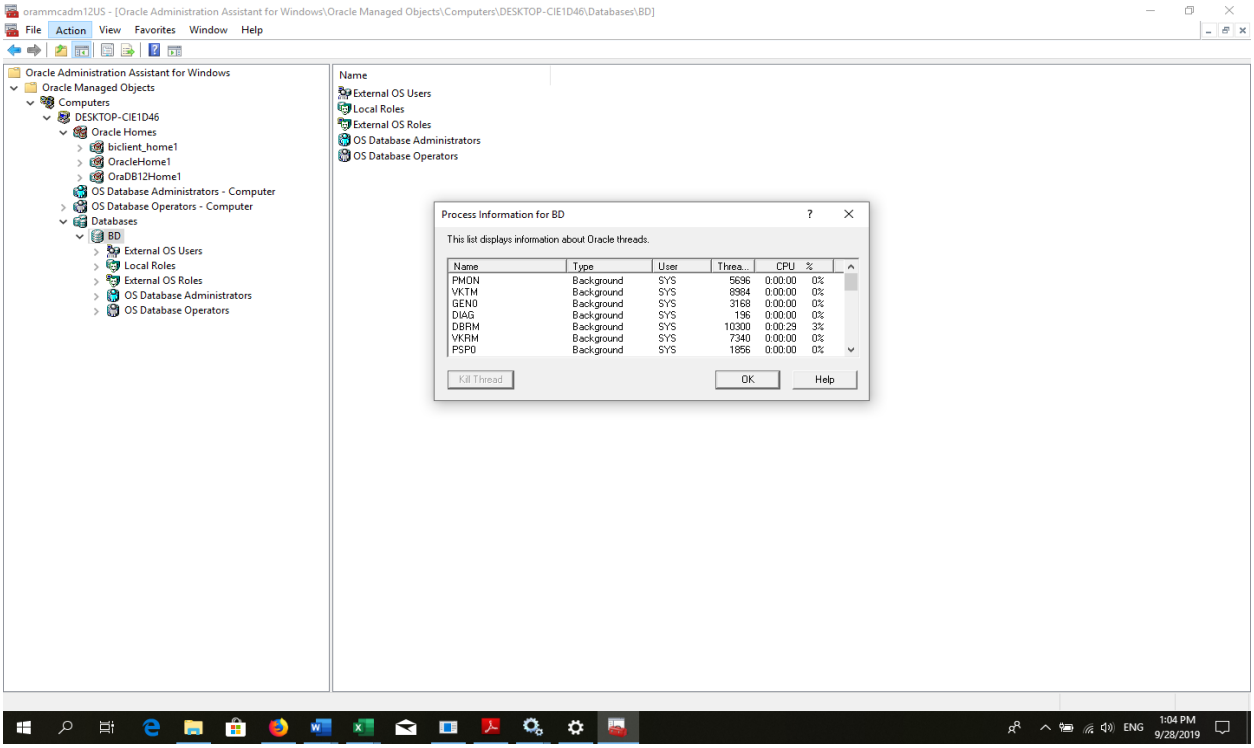
E. Utilitarul Oracle Administration Assistant



1. Parametrii de Startup/Shutdown:



2. Informatii despre procese:



3. Modificare credentiale pentru userii externi:

